Homework #1 Solutions

6 Points Completion

Chapter 0: 7 c,d; 8 c,d; 32, 43 a,c; 44, 46, 51, 52

7 c,d)

(1 Point) (c) 20000+12FF = 212FF
(d) FFFF+2222 = 12221

8 c,d)

(**2 Points**) (c) 2FFFF-FFFFF = 30000h
(d) 9FF25-4DD99 = 5218Ch

32) True

(1 Point) 43) (a) 128K 14 8

(c) 512K 16 8

(1 Point) 44) 4000h - 7FFFh

(2 Points) 46) Y0=> 8000h - 8FFFh, Y3 => B000h - BFFFh, Y6=> E000h – EFFFh

(1 Point) 51. PC (Program Counter)
52. ALU is responsible for all arithmetic and logic calculations in the CPU.

Chapter 1: 2, 3, 4, 11-13

(1 Point) 1. False.
2. True.
3. True.
4. True.

(1 Point) 11. Power consumption.
12. The ROM area is where the executable code is stored.
13. Very, in case there is a shortage by one supplier.

PROBLEMS

SECTION 0.1: NUMBERING AND CODING SYSTEMS

1. Convert the following decimal numbers to binary:
   (a) 12 (b) 123 (c) 63 (d) 128 (e) 1000
2. Convert the following binary numbers to decimal:
   (a) 100100 (b) 1000001 (c) 11101 (d) 1010 (e) 00100010
3. Convert the values in Problem 2 to hexadecimal.
4. Convert the following hex numbers to binary and decimal:
   (a) 2B9H (b) F44H (c) 912H (d) 2BH (e) FFFFH
5. Convert the values in Problem 1 to hex.
6. Find the 2's complement of the following binary numbers:
   (a) 1001010 (b) 111001 (c) 10000010 (d) 111110001
7. Add the following hex values:
   (a) 2CH + 3FH (b) F34H + 5D6H (c) 20000H + 12FFH
   (d) FFFFH + 2222H
8. Perform hex subtraction for the following:
   (a) 24FH – 129H (b) FE9H – 5CCH (c) 2FFFFH – FFFFFH
   (d) 9FF25H – 4DD99H
9. Show the ASCII codes for numbers 0, 1, 2, 3, ..., 9 in both hex and binary.
10. Show the ASCII code (in hex) for the following strings:
    "U.S.A. is a country" CR,LF
    "in North America" CR,LF
    (CR is carriage return, LF is line feed)

SECTION 0.2: DIGITAL PRIMER

11. Draw a 3-input OR gate using a 2-input OR gate.
12. Show the truth table for a 3-input OR gate.
13. Draw a 3-input AND gate using a 2-input AND gate.
14. Show the truth table for a 3-input AND gate.

28. What is the difference in capacity between a 4M memory chip and 4M of computer memory?
29. True or false. The more address pins, the more memory locations are inside the chip. (Assume that the number of data pins is fixed.)
30. True or false. The more data pins, the more each location inside the chip will hold.
31. True or false. The more data pins, the higher the capacity of the memory chip.
32. True or false. The more data pins and address pins, the greater the capacity of the memory chip.
33. The speed of a memory chip is referred to as its _____.
34. True or false. The price of memory chips varies according to capacity and speed.
35. The main advantage of EEPROM over UV-EPROM is _____.
36. True or false. SRAM has a larger cell size than DRAM.
37. Which of the following, EPROM, DRAM, or SRAM, must be refreshed periodically?
38. Which memory is used for PC cache?
39. Which of the following, SRAM, UV-EPROM, NV-RAM, or DRAM, is volatile memory?
40. RAS and CAS are associated with which type of memory?
    (a) EPROM (b) SRAM (c) DRAM (d) all of the above
41. Which type of memory needs an external multiplexer?
    (a) EPROM (b) SRAM (c) DRAM (d) all of the above
42. Find the organization and capacity of memory chips with the following pins.
    (a) EEPROM A0–A14, D0–D7
    (b) UV-EPROM A0–A12, D0–D7
    (c) SRAM A0–A11, D0–D7
    (d) SRAM A0–A12, D0–D7
    (e) DRAM A0–A10, D0
    (f) SRAM A0–A12, D0
    (g) EEPROM A0–A11, D0–D7
    (h) UV-EPROM A0–A10, D0–D7
    (i) DRAM A0–A8, D0–D3
    (j) DRAM A0–A7, D0–D7
43. Find the capacity, address, and data pins for the following memory organizations.
    (a) 16K × 8 ROM
    (b) 32K × 8 ROM
    (c) 64K × 8 SRAM
    (d) 256K × 8 EEPROM
    (e) 64K × 8 ROM
    (f) 64K × 4 DRAM
    (g) 1M × 8 SRAM
    (h) 4M × 4 DRAM
    (i) 64K × 8 NV-RAM

44. Find the address range of the memory design in the diagram.
45. Using NAND gates and inverters, design decoding circuitry for the address range 2000H–2FFFH.
46. Find the address range for Y0, Y3, and Y6 of the 74LS138 for the diagrammed

design.

47. Using the 74138, design the memory decoding circuitry in which the memory block controlled by Y0 is in the range 0000H to 1FFFH. Indicate the size of the memory block controlled by each Y.
48. Find the address range for Y3, Y6, and Y7 in Problem 47.

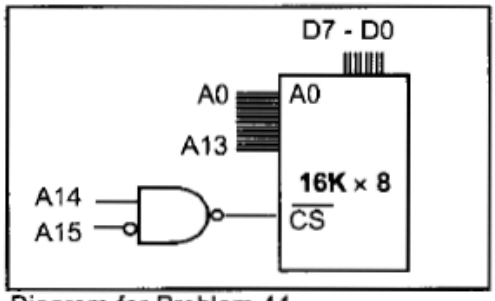


Diagram for Problem 44

49. Using the 74138, design memory decoding circuitry in which the memory block controlled by Y0 is in the 0000H to 3FFFH space. Indicate the size of the memory block controlled by each Y.
50. Find the address range for Y1, Y2, and Y3 in Problem 49.

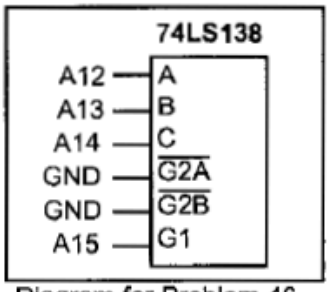


Diagram for Problem 46

SECTION 0.4: CPU AND HARVARD ARCHITECTURE

51. Which register of the CPU holds the address of the instruction to be fetched?
52. Which section of the CPU is responsible for performing addition?
53. List the three bus types present in every CPU.

SECTION 1.1: MICROCONTROLLERS AND EMBEDDED PROCESSORS

1. True or False. A general-purpose microprocessor has on-chip ROM.
2. True or False. Generally, a microcontroller has on-chip ROM.
3. True or False. A microcontroller has on-chip I/O ports.
4. True or False. A microcontroller has a fixed amount of RAM on the chip.
5. What components are usually put together with the microcontroller onto a single chip?
6. Intel's Pentium chips used in Windows PCs need external _____ and _____ chips to store data and code.
7. List three embedded products attached to a PC.
8. Why would someone want to use an x86 as an embedded processor?
9. Give the name and the manufacturer of some of the most widely used 8-bit microcontrollers.
10. In Question 9, which one has the most manufacture sources?
11. In a battery-based embedded product, what is the most important factor in choosing a microcontroller?
12. In an embedded controller with on-chip ROM, why does the size of the ROM matter?
13. In choosing a microcontroller, how important is it to have multiple sources for that chip?
14. What does the term "third-party support" mean?
15. Suppose that a microcontroller architecture has both 8-bit and 16-bit versions. Which of the following statements is true?
    (a) The 8-bit software will run on the 16-bit system.
    (b) The 16-bit software will run on the 8-bit system.